

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

**Duration : 1 Hour
Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I Beffy A Shanika(192511387) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

beffy

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q1.Backtracking Search

Given Constraints

- D1 cannot work Night
- D2 must work before D3 (Morning < Afternoon < Night)
- One doctor per shift
- D3 cannot work Morning
- Every doctor gets exactly one shift

Step 1: Assign D1

Try Morning

Current Assignment:

- Morning → D1

Step 2: Assign D2

Try Afternoon

Current Assignment:

- Morning → D1
- Afternoon → D2

Step 3: Assign D3

Only Night remains.

Check constraints:

- D3 not Morning
- D2 before D3 (Afternoon before Night)
- One doctor per shift
- D1 not Night

All constraints satisfied.

Backtracking Search Tree

Start

```
|
|-- D1 → Morning
|  |
|  |-- D2 → Morning ✗ (Shift occupied)
|  |
|  |-- D2 → Afternoon
|      |
|      |-- D3 → Morning ✗
|      |
|      |-- D3 → Night ✓
```

Final Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Q2. BFS Search Algorithm

Assume the grid is:

```
S . . X .
. X . X .
. X . . .
. . X X .
. . . G .
```

S = Start

G = Goal

X = Obstacle

Manhattan Distance

$$h = |x_1 - x_2| + |y_1 - y_2|$$

BFS Node Expansion

Queue = [S]

Expand S

Queue:

[(1,0),(0,1)]

Expand (1,0)

Queue:

[(0,1),(2,0)]

Expand (0,1)

Queue:

[(2,0),(0,2)]

Expand (2,0)

Queue:

[(0,2),(3,0)]

Continue expanding level by level until Goal is reached.

Optimal Path

S

↓

↓

↓

→

→

↓

→

G

Example Path:

(0,0)

↓

(1,0)

↓

(2,0)

↓

(3,0)

→

(3,1)

↓

(4,1)

→

(4,2)

→

(4,3)=Goal

Cost

Each move = 1

Total Moves = 7

Optimal Cost = 7

Q3. Rescue Robot Problem

(a) Constraint Analysis

- Robot moves only Up, Down, Left, Right.
- Obstacles cannot be crossed.
- Limited sensing means it knows only nearby cells.
- Normal move cost = 1.
- Risky cell adds extra cost (+2).
- Robot updates map while moving (Online Search).
- Objective: Reach survivor safely with minimum cost.

(b) Solution Using Uniform Cost Search (UCS) / Online Search

Steps

1. Start from S.
2. Observe neighbouring cells.
3. Ignore blocked cells.
4. Calculate movement cost.
5. Add risky zone cost if present.
6. Expand the node with the lowest total cost.
7. Update knowledge whenever new obstacles are found.
8. Continue until Goal is reached.

(c) AI Concepts

Intelligent Agent

The rescue robot senses its environment and performs actions to reach the survivor.

Rationality

The robot always chooses the path with the least total cost while avoiding danger.

Environment Type

- Partially Observable
- Dynamic
- Sequential
- Discrete
- Single Agent

(d) Justification

Uniform Cost Search (UCS) is the most suitable because:

- Finds the minimum-cost path.
- Considers risky zones.
- Works correctly when movement costs differ.
- Guarantees the optimal solution.
- Easily adapts to online updates in a partially observable environment.